

Naive Bayes Classifier

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Bayes' Rule

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where H and E are events and $P(E) \neq 0$.

Terminology

- ▶ $P(H | E)$: **The probability of hypothesis H , given evidence E .**
- ▶ $P(E)$: **Marginal probability of the evidence E .**
- ▶ $P(H)$: **Prior probability of hypothesis H .**
- ▶ $P(E | H)$: **Likelihood of the evidence given hypothesis.**
- ▶ $P(H | E)$: **Posterior probability of hypothesis H .**

Bayes' Rule

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where H and E are events and $P(E) \neq 0$

Bayes' Rule is useful for estimating $P(H | E)$ when it is hard to estimate $P(H | E)$ directly from the training data, but $P(E | H)$, $P(H)$, and $P(E)$ can be estimated more easily.

Bayes' Rule: derivation

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where H and E are events and $P(E) \neq 0$

By the definition of conditional probability, we have

$$P(H | E) = \frac{P(H, E)}{P(E)} \quad \text{and} \quad P(E | H) = \frac{P(H, E)}{P(H)},$$

from which we derive Bayes' Rule.

Example (single feature)

- ▶ **Meningitis** causes a **stiff neck** 50% of the time.
- ▶ Meningitis occurs 1/50000 and stiff neck occurs 1/20.
- ▶ Compute the probability of meningitis, given that the patient has a stiff neck.
- ▶ H = meningitis, E = stiff neck
- ▶ $P(H) = 1/50000$, $P(E) = 1/20$, $P(E | H) = 0.5$
- ▶ From Bayes' rule we have

$$P(H | E) = \frac{P(E | H) P(H)}{P(E)} = 0.0002$$

Example

- ▶ **Meningitis** causes a **stiff neck** 50% of the time.
- ▶ Meningitis occurs $1/50000$ and stiff neck occurs $1/20$.
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- ▶ H = meningitis, E = stiff neck
- ▶ $P(H) = 1/50000$, $P(E) = 1/20$, $P(E | H) = 0.5$
- ▶ From Bayes' rule we have

$$P(H | E) = \frac{P(E | H) P(H)}{P(E)} = 0.0002$$

- ▶ If we have 1-dimensional space (only one feature: $\bar{X} = (a)$), then we can estimate $P(H | \bar{X})$ directly from the training data set.
- ▶ It becomes more problematic if we have higher dimension. Say $\bar{X} = (a_1, a_2, \dots, a_d)$ for $d > 1$.

Example (2 features)

- ▶ Let H = engine-does-not-start
- ▶ Evidences A = weak-battery and B = no-gas
- ▶ To estimate $P(H | A, B)$ directly, we need to restrict our consideration only to those cars (objects) in the dataset that had a **weak battery** and **no gas**. Among those we need to count the cars with **non-working engine**. Such cases could be rare in our dataset making estimate of $P(H | A, B)$ unreliable or zero (in the worst case).

$$P(H | A, B) = \frac{\# \text{ weak-bat. \& no-gas \& eng.-not-working}}{\# \text{ weak-bat. \& no-gas}}$$

- ▶ Bayes' rule provides a way of expressing $P(H | A, B)$ directly in terms of $P(A, B | H)$:

$$P(H | A, B) = \frac{P(A, B | H)P(H)}{P(A, B)}$$

and estimate the latter using **naive Bayes approximation** (which is much easier to do).

Naive Bayes approximation

- ▶ Let C be a random variable representing the class of an unseen d -dimensional test object $\bar{X} = (x_1, x_2, \dots, x_d)$, where $x_1, x_2, \dots, x_d$ denote random variables of individual dimensions.
- ▶ Given a specific test object $(a_1, a_2, \dots, a_d)$ the goal is to estimate

$$P(C = c \mid \bar{X} = (a_1, a_2, \dots, a_d)) = P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)$$

- ▶ By Bayes' rule

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C=c) P(x_1=a_1, x_2=a_2, \dots, x_d=a_d \mid C)}{P(x_1=a_1, x_2=a_2, \dots, x_d=a_d)}$$

Naive Bayes approximation

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)}{P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)}$$

Does not depend on the class variable C

The class c with the largest numerator

$$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

has the largest **posterior** probability

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)$$

Naive Bayes approximation

$$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

- ▶ $P(C = c)$: Can be **estimated** as the fraction of the training data objects that belong to class c .
- ▶ How to estimate $P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$?

Naive assumption

The values of different features $x_1, x_2, \dots, x_d$ are **independent** of one another **conditional on the class**.

Independent events (2 events)

Joint probability of two events A and B

$$P(A, B) = P(A | B)P(B)$$

this holds for ANY pair of events A and B , irrespective of whether they are independent or not.

If A is independent of B , then B 's occurrence has no “consequence” on A

$$P(A | B) = P(A)$$

Thus, when A and B are **independent**, their joint probability

$$P(A, B) = P(A)P(B)$$

Independent events (> 2 events)

A finite set $\{A_1, A_2, \dots, A_n\}$ of events is **mutually independent** if for any $2 \leq k \leq n$ and for any subset of k events

$$\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\} \subseteq \{A_1, A_2, \dots, A_n\}$$

we have

$$P(A_{i_1}, A_{i_2}, \dots, A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}) = \prod_{j=1}^k P(A_{i_j})$$

Naive Bayes approximation

How to estimate $P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$?

Naive assumption

The values of different features $x_1, x_2, \dots, x_d$ are **independent** of one another **conditional on the class**.

$$P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c) = \prod_{i=1}^d P(x_i = a_i \mid C = c)$$

$P(x_i = a_i \mid C = c)$: Can be estimated as the fraction of training objects in class c that have feature $x_i = a_i$.

Naive Bayes approximation

$P(x_i = a_i \mid C = c)$:

Can be estimated as the fraction of training objects in class c that have feature $x_i = a_i$

This is much easier to estimate from the training data than

$P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$

Hence, under the independence assumption, the estimation of

$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c) = P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$

reduces to the estimations of

$P(C), P(x_1 = a_1 \mid C = c), P(x_2 = a_2 \mid C = c), \dots, P(x_d = a_d \mid C = c)$

Being naive makes life easy

Example (2 features)

- ▶ Let H = engine-does-not-start
- ▶ Evidences A = weak-battery and B = no-gas
- ▶ Direct estimation

$$P(H | A, B) = \frac{\# \text{ weak-bat. \& no-gas \& eng.-not-working}}{\# \text{ weak-bat. \& no-gas}}$$

- ▶ Using Bayes' rule + **Naive Bayes approximation**

$$P(H | A, B) = \frac{P(A, B | H)P(H)}{P(A, B)} = \frac{P(A | H)P(B | H)P(H)}{P(A, B)}$$

- ▶ $P(A | H)$ can be estimated as the fraction of cars with **weak battery** among cars with **engine not working**
- ▶ $P(B | H)$ can be estimated as the fraction of cars with **no gas** among cars with **engines not working**

Making the independence assumption makes estimates possible in practice

Bayes' rule: Proportional Form

- ▶ Assume $\overline{X} = (a_1, a_2, \dots, a_d)$ is the input test object
- ▶ We need to select/predict the class of $\overline{X}$ from the set $\{c_1, c_2, \dots, c_k\}$.

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)}{P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)}$$

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) \propto P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

$$P(C \mid X) \propto P(C) P(X \mid C)$$

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X}$ = (Outlook = sunny, Temp = cool, Humidity = high, Windy = true)

$$\begin{aligned}
 P(\text{Play}=\text{yes}|\bar{X}) &\propto P(\bar{X}|\text{Play}=\text{yes})P(\text{Play}=\text{yes}) \\
 &= P(\text{sunny}|\text{yes})P(\text{cool}|\text{yes})P(\text{high}|\text{yes})P(\text{true}|\text{yes})P(\text{yes}) \\
 &= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529
 \end{aligned}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X} = (\text{Outlook} = \text{sunny}, \text{Temp} = \text{cool}, \text{Humidity} = \text{high}, \text{Windy} = \text{true})$

$$\begin{aligned}P(\text{Play}=\text{no}|\bar{X}) &\propto P(\bar{X}|\text{Play}=\text{no})P(\text{Play}=\text{no}) \\&= P(\text{sunny}|\text{no})P(\text{cool}|\text{no})P(\text{high}|\text{no})P(\text{true}|\text{no})P(\text{no}) \\&= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.020\end{aligned}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X}$ = (Outlook = sunny, Temp = cool, Humidity = high, Windy = true)

Posterior comparison

$$P(\text{Play=yes}|\bar{X}) \propto P(\bar{X}|\text{Play=yes})P(\text{Play=yes}) = 0.00529$$

$$P(\text{Play=no}|\bar{X}) \propto P(\bar{X}|\text{Play=no})P(\text{Play=no}) = 0.020$$

Therefore, Play = no

Naive Bayes: classification task

In the previous setting (**choose the most probable class**):

- ▶ Given a test object $\bar{X} = (a_1, a_2, \dots, a_d)$, we wanted to predict its class C
- ▶ For this, we used the proportional form

$$P(C = c \mid X = \bar{X}) \propto P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$$

and it was enough to find $c \in \{c_1, c_2, \dots, c_k\}$ that maximises

$$P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$$